

Non-Linear Dynamics Notes

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Lecture 1

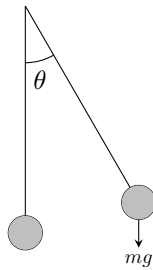
1.1 Books

- Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed., Westview Press, 2015.

This is Bible for the subject. It is standard, detailed and a bit old text.

1.2 Dynamics

- Study of evolution of systems
Eg. Simple Pendulum



has equation of motion

$$\ddot{\theta} - \frac{g}{l} \sin \theta = 0 \quad (1.1)$$

For small angles

$$\ddot{\theta} - \frac{g}{l} \theta = 0 \quad (1.2)$$

Which is linear. But if θ is not small then non-linear and hence included in NLD¹.

- As equation becomes Non-Linear, the solution drastically changes

¹**NLD**: Non-Linear Dynamics

- Non-Negligible Non-Linear terms are not perturbative and needs separate study
- The subject NLD has advanced after the computers became powerful enough to perform high level computation

1.3 1D Maps

- Suppose we have a sequence of numbers (real numbers mostly)

$$x_0, x_1, x_2, \dots$$

Such that

$$x_0, x_1 = f(x_0), x_2 = f(x_1), \dots \quad (1.3)$$

i.e.

$$x_{n+1} = f(x_n) \quad (1.4)$$

Then it is called an *Map*²³

1.3.1 Fixed Point

- Solution of (1.3) such that

$$x^* = f(x^*) \quad (1.5)$$

then x^* ⁴ is called *fixed point*. Note that it will give rise to the sequence of form

$$x_0, x_1, x_2, \dots, x^*, x^*, x^*, \dots$$

1.3.2 Stability of a Fixed Point

- Suppose $x_0 = \bar{x} + \delta$.

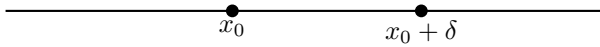


Fig. 1

- Now if the subsequent x 's approach the fixed point x^* then the fixed point is stable and called an *attractor*

²Note here that the function f remains same

³If the function f has random number in the function itself then it is *Undeterministic Map*. We'll consider only *Deterministic Maps*

⁴It is widely followed conventions to label fixed point as x^* and we'll continue using it

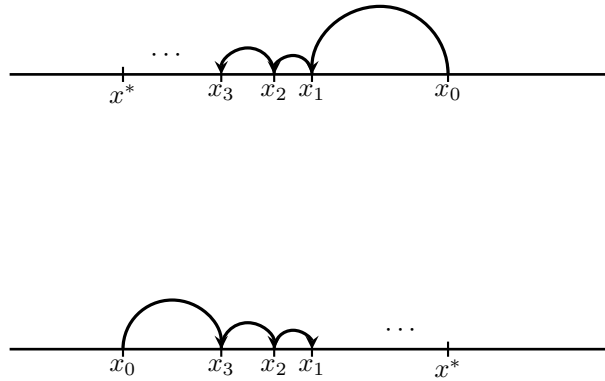


Fig. 2

- Approach to x^* is Oscillatory and is still fixed point

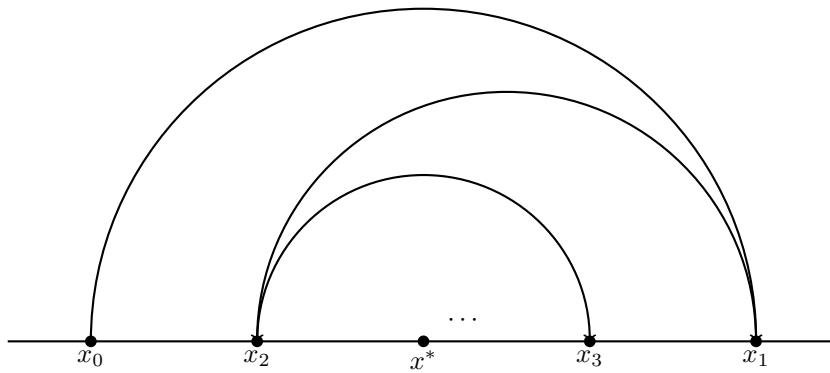


Fig. 3

- If the subsequent points diverge from the fixed point (monotonically or oscillatory) then the point x^* is Unstable Fixed Point

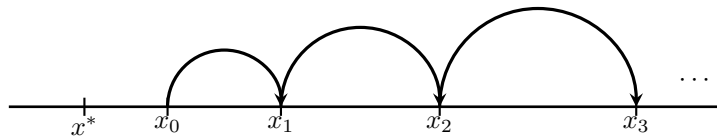


Fig. 4

1.3.3 Digramatic way to find number of fixed points

Follow the steps:

1. Draw the line $y = x$
2. Draw the curve $y = f(x)$
3. No. of intersections = No. of Fixed Points⁵

Example 1. $f(x) = \sin x$

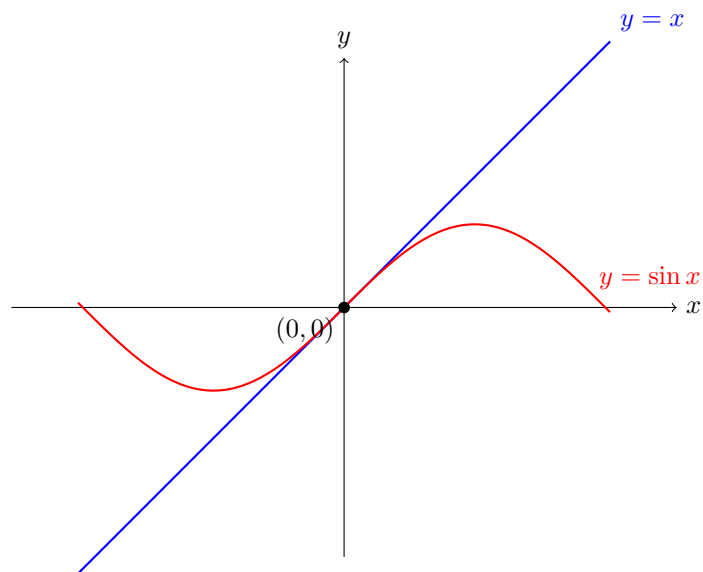
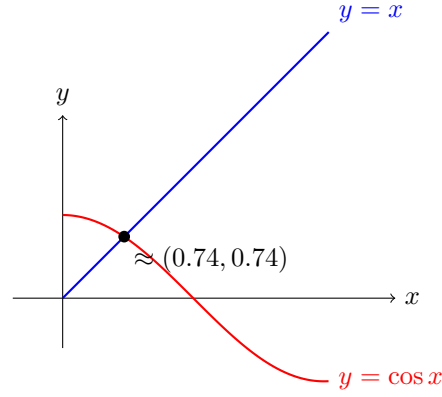


Fig. 5

There is only one fixed point at $x = 0$

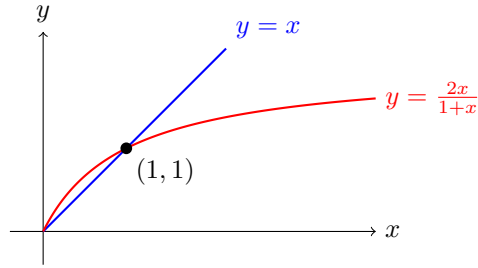
Example 2. $f(x) = \cos x$

⁵This logic is clear from the definition of the fixed point as (1.3.1)

**Fig. 6**

There is only one fixed point at $x \approx 0.74$

Example 3. $f(x) = \frac{2x}{1+x}$

**Fig. 7**

There is only one fixed point at $x = 1$

- In monotonic approach
 $|x_{n+1} - x^*| < |x_n - x^*| \implies$ Stable fixed point
 $|x_{n+1} - x^*| > |x_n - x^*| \implies$ Unstable fixed point
- We define the quantity *multiplier* by

$$\mu = \frac{x_1 - x^*}{x_0 - x^*} \quad (1.6)$$

It determines whether the fixed point is stable or unstable.

$$|\mu| < 1 \implies \text{Stable} \quad (1.7)$$

$$|\mu| > 1 \implies \text{Unstable} \quad (1.8)$$

$$\mu > 0 \implies \text{Monotonic} \quad (1.9)$$

$$\mu < 0 \implies \text{Oscillatory} \quad (1.10)$$

- Now, let's analyze the fixed point analytically. Let

$$x_0 = x^* + \delta \quad (1.11)$$

Then

$$\begin{aligned} x_1 &= f(x_0) = f(x^* + \delta) \\ &= f(x^*) + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \dots \\ &= x^* + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \dots \end{aligned}$$

- Now linearize

$$x_1 = x^* + \delta f'(x^*) \quad (1.12)$$

Therefore,

$$f'(x^*) = \frac{x_1 - x^*}{\delta} = \frac{x_1 - x^*}{x_0 - x^*}$$

Using (1.6)

$$\mu = f'(x^*) \quad (1.13)$$

- Now in Example 1: $\mu = \cos x^* = 1$. The case $\mu = 1$ ⁶ needs special analysis and we skip it for a moment
- In Example 2: $\mu = -\sin(x^*) < 0$, hence fixed point is stable and oscillatory.
- In Example 3:

$$\mu = \frac{2(1+x^*) - 2x^*}{(1+x^*)^2} = \frac{2}{(1+x^*)^2}$$

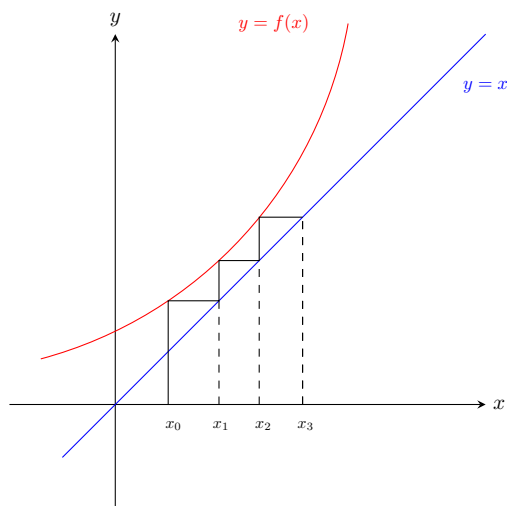
At $x^* = 0$; $\mu = 2 \implies$ monotonically unstable

At $x^* = 1$; $\mu = 1/2 \implies$ monotonically stable

- Calculating Stability form multiplier is only valid when Linearization is valid.
- We now adopt Diagramatic Approach: *Cobwebb Diagrams*
 1. Draw curve $y = x$
 2. Draw $y = f(x)$

⁶In this case the fixed point x^* is called *Marginally Stable*

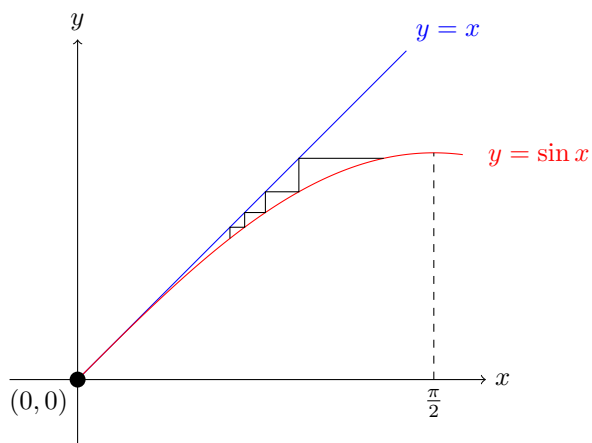
3. Draw line from x_0 to $f(x)$
4. Draw horizontal line to $y = x$ and that point of intersection gives x_1

**Fig. 8**

The figure above depicts an example.

Now let's draw Cobweb diagrams to our three examples.

- For Example 1

**Fig. 9**

Note that Cobwebb diagram shows $x = 0$ as stable fixed point unlike our analytical approach which failed due to $\mu = 1$.

- For Example 2

$\mu = 0$ and $\mu = 1$ **Case:**

$$x_1 = x^* + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \dots$$

- for $\mu = 1$

$$\begin{aligned} x_1 &= x^* + \delta + \frac{1}{2}\delta^2 f''(x^*) + \dots \\ &= x_0 + \frac{1}{2}\delta^2 f''(x^*) + \dots \end{aligned} \quad (1.14)$$

hence from above the higher order terms are important and will determine stability.

- For $\mu = 0$

$$\begin{aligned} x_1 &= x^* + \frac{1}{2}\delta^2 f''(x^*) + \dots \\ &\approx x^* + c\delta^2 \quad \left\{ c = \frac{1}{2}f''(x^*) \right\} \end{aligned} \quad (1.15)$$

Now,

$$x_2 = x^* + c(c\delta^2)$$

And hence,

$$x_n = x^* + c^{(\dots)}\delta^{2^n} \quad (1.16)$$

Since convergence is very fast, it is *superstable* fixed point.

-

$$\begin{aligned} x_1 &= f(x_0) = f(x^* + \delta) \\ &= x^* + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \dots \end{aligned}$$

Linearize

$$\begin{aligned} x_1 &= x^* + \delta\mu \\ x_2 &= f(x_1) = f(x^* + \delta\mu) \\ &= x^* + \mu^2\delta \\ x_3 &= \mu^3\delta \\ x_n &= x^* + \mu^3\delta \end{aligned} \quad (1.17)$$

$$\begin{aligned} \frac{x_n - x^*}{x_0 - x^*} &= \mu^n = e^{n \log \mu} \quad (\mu > 0) \\ &= (|\mu| \text{sign}(\mu))^n \\ &= [\text{sign}(\mu)]^n e^{n \log |\mu|} \\ x_n &= x^* + \delta \cdot (\text{sign}(\mu))^n e^{n \log |\mu|} \end{aligned} \quad (1.18)$$

1.4 Logistic Map

We define *Logistic Map*⁷⁸ by

$$f(x) = rx(1 - x) \quad r > 0 \quad (1.19)$$

- Now let's find the fixed points

$$x^* = rx^*(1 - x^*)$$

$$\boxed{x^* = 0} \quad (1.20)$$

$$1 = r(1 - x^*)$$

$$\boxed{x^* = 1 - \frac{1}{r}} \quad (1.21)$$

- We'll always keep $0 \leq |x_n| \leq 1$ and vary $r > 0$
 $x^* = 0$ exist for all r
 $x^* = 1 - \frac{1}{r}$ exist for $r \geq 0$
- Stability of Fixed Points

$$f'(x) = r - 2rx$$

At $x^* = 0$ $f(x^*) = r \implies$ stable fixed point for $r < 1$ and unstable for $r > 1$; $r = 1$ is marginally stable
 At $x^* = 1 - \frac{1}{r}$

$$\begin{aligned} f(x^*) &= r - 2r \left(1 - \frac{1}{r}\right) \\ &= r - 2r + 1 \\ &= 2 - r \end{aligned}$$

for $r = 1$ $\mu = -1$ marginally stable
 for $r = 3$ $\mu = 1$ marginally stable

- for $r > 3$ (say $r = 3.1$) we get sequence

$$x_0, x_1, \dots, p, q, p, q, \dots \quad (1.22)$$

This alternate sequence of p 's and q 's is called a *2-cycle*⁹.

⁷ $f(x)$ is quadratic hence included in NLD

⁸This equations appears in many areas: Engineernig, Population Dynamics, etc.

⁹A fixed point is a 1-cycle

Lecture 2

2.1 Logistic Map (cont...)

- For stability

$$\begin{aligned} -1 < \mu < 1 &\implies -1 < 2 - r < 1 \\ &\implies -3 < -r < -1 \\ &\implies 1 < r < 3 \end{aligned}$$

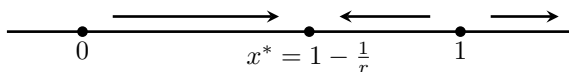


Fig. 2.1

- for a 2-cycle

$$\begin{aligned} f(p) &= q & f(q) &= p \\ \therefore f(f(x)) &= f(x) = x \end{aligned} \tag{2.1}$$

and solution must be p and q

$$\begin{aligned} r(rx(1-x))(1-rx(1-x)) &= x \\ r^2x(1-x)(1-rx(1-x)) &= x \end{aligned}$$

$f(x^*) = x^*$ and therefore $f(f(x^*)) = f(x^*) = x^*$

Therefore, $0, 1 - \frac{1}{r}$ are solution of above equation

$$x [r^2(1-x)(1-rx(1-x)-1)] = 0$$

taking out $x = 0$ factor

$$r^2(1-x)(1-rx(1-x)-1) = 0$$

$$\text{put } r(1-x) = y \implies rx = r-y$$

$$ry(1-xy) - 1 = 0$$

$$yr - y^2rx - 1 = 0$$

$$yr - y^2(r-y) - 1 = 0$$

$$(y-1)(-yr + y^2 + y + 1) = 0$$

$$y = 1 \implies r(1-x) = 1 \implies x = 1 - \frac{1}{r}$$

$$y^2 + y(1-r) + 1 = 0 \quad (2.2)$$

Now, consider

$$\begin{aligned} y &= \frac{-(1-r) \pm \sqrt{(1-r)^2 - 4}}{2} \\ &= \frac{(r-1) \pm \sqrt{r^2 - 2r - 3}}{2} \\ &= \frac{(r-1) \pm \sqrt{(r-3)(r+1)}}{2} \\ r(1-x) &= \frac{(r-1) \pm \sqrt{(r-3)(r+1)}}{2} \\ x &= 1 - \frac{(r-1) \pm \sqrt{(r-3)(r+1)}}{2r} \end{aligned}$$

$$x = \frac{1}{2r} \left(r+1 \pm \sqrt{(r-3)(r+1)} \right) \quad (2.3)$$

•

$$p+q = \frac{r+1}{r} = 1 + \frac{1}{r} \quad (2.4)$$

$$\begin{aligned} pq &= \frac{1}{4r^2} \left[(r+1)^2 - (r+1)(r-3) \right] \\ &= \frac{1}{4r^2} (r^2 + 2r + 1 - r^2 + 2r + 3) \\ &= \frac{1}{4r^2} (4r + 4) \end{aligned}$$

$$pq = \frac{1}{r} \left(1 + \frac{1}{r} \right) \quad (2.5)$$

- The 2-cycle exists only for $r > 3$
 $x^* = 1 - \frac{1}{r}$ stable for $1 < r < 3$

- Suppose we start with the point $p + \delta_0$, then we'll get sequence

$$p + \delta_0, p + \delta_1, p + \delta_2, \dots$$

Question: Does $\delta_n \rightarrow 0$

$$\begin{aligned} \mu_2 = \left| \frac{x_2 - p}{x_0 - p} \right| < 1 &\implies \text{We approach gradually to } p; \text{ stable 2-cycle} \\ &> 1 \implies \text{unstable 2-cycle} \end{aligned}$$

•

$$\begin{aligned} x_2 &= f^2(x_0) \\ &= f^2(p + \delta) \\ &= f^2(p) + \delta \frac{d^2}{dx^2} f^2(p) \Big|_{x=p} \end{aligned}$$

Write $g = f(x) \implies g(p) = f(p) = p$ and $f^2(x) = f(g)$

$$\begin{aligned} \mu_2 &= \frac{d^2}{df(x)^2} f^2(x) \frac{df(x)}{dx} \Big|_{x=p} \\ \mu_2 &= \frac{df(g)}{dg} \frac{df(x)}{dx} \Big|_{x=p} \end{aligned} \tag{2.6}$$

$$\begin{aligned} \mu_2 &= (r - 2rg)(r - 2rx) \\ &= r^2(1 - 2g)(1 - 2x) \Big|_{x=p} \end{aligned} \tag{2.7}$$

let $g = f(x) \quad q = f(p)$

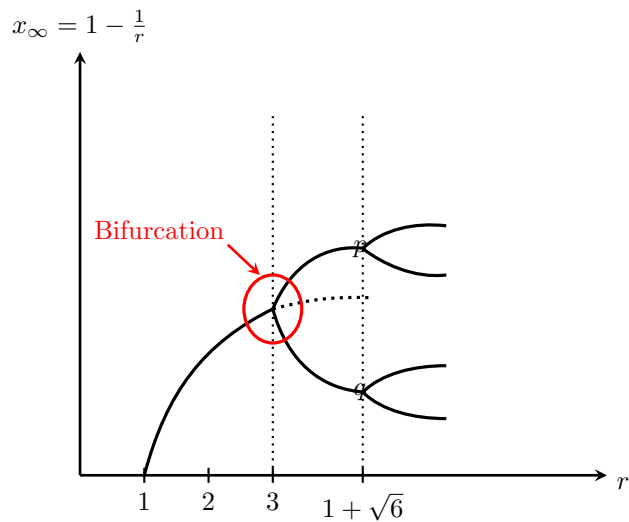
$$\begin{aligned} \mu_2 &= r^2(1 - 2q)(1 - 2p) \\ &= r^2 \left[1 - 2 \left(1 + \frac{1}{r} \right) + \frac{4}{r} \left(1 + \frac{1}{r} \right) \right] \\ &= r^2 \left(-1 + \frac{2}{r} + \frac{4}{r^2} \right) \\ &= 4 + 2r + r^2 \\ &= 5 - (r^2 - 2r + 1) \end{aligned}$$

$$\mu_2 = 5 - (r - 1)^2 \tag{2.8}$$

$|\mu_2| < 1$ (Condition for stability of 2-cycle)

$$\begin{aligned} &\implies -1 < \mu_2 < 1 \\ &\implies -1 < 5 - (r - 1)^2 < 1 \\ &\implies -6 < -(r - 1)^2 < -4 \\ &\implies 4 < (r - 1)^2 < 6 \end{aligned}$$

$$\boxed{3 < r < 1 + \sqrt{6}} \quad (2.9)$$



- **Bifurcation:** A fixed point splits in two¹
- $1 - \frac{1}{r}$ exists for $r \geq 1$ but stable for $1 < r < 3$
2-cycle p, q exists for $r > 3$ but stable for $3 < r < 1 + \sqrt{6}$
- for $r > 1 + \sqrt{6}$ we get

$$x_0, x_1, x_2, \dots, p, q, r, s, p, q, r, s, \dots \quad (2.10)$$

which is a 4-cycle²

¹Refer to *H. G. Schuster and W. Just, *Deterministic Chaos*, 3rd ed., Wiley-VCH* for nice diagrams

²As my professor said promotion from bicycle to motorcycle